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BUILD
DEPLOY
INNOVATE
GROW

Cloud Computing with AWS

Learn. Build. Deploy. Scale.

Real-World Cloud Skills for a Global Career.

- Learn Cloud Skills
- Hands-on Projects
- Real-World Use Cases
- Build Your Future

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What is Cloud Computing?

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, software and more over the internet instead of owning and managing physical hardware.

In simple terms, cloud computing allows you to access and use technology resources on demand, over the internet, and pay only for what you use.

Key Benefits

Scalability
Increase or decrease resources easily

Cost Savings
Pay only for what you use

Flexibility
Access resources anytime, anywhere

Reliability
High availability and disaster recovery

Focus on Innovation
Spend less time on infrastructure and more time on what matters

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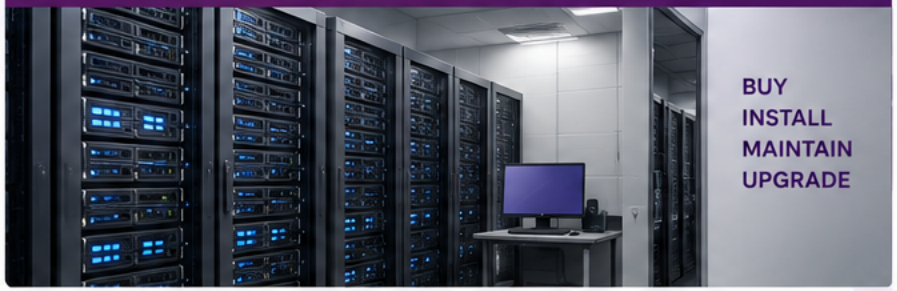
SAME GOAL.
A BRIGHTER
FUTURE.

Traditional IT Infrastructure VS Cloud Computing

DIFFERENT APPROACHES. SAME POSSIBILITIES.

Traditional IT Infrastructure

On-Premises. You Own and Manage.



Cloud Computing (AWS)

On-Demand. You Use and Pay for What You Need.

Traditional IT Infrastructure		Cloud Computing (AWS)
High upfront cost (servers, storage, networking, etc.)	**Cost**	Pay only for what you use (no large upfront cost)
Physical hardware to install and maintain	**Infrastructure**	Virtual infrastructure in the cloud (no physical hardware)
Limited and slow to scale (requires new hardware)	**Scalability**	Easily scale up or down in minutes
You manage power, cooling, security, and maintenance	**Management**	Managed by cloud provider (AWS handles infrastructure)
Hardware and software upgrades can take time	**Deployment Speed**	Provision resources in minutes
Access mainly from office / on-site network	**Accessibility**	Access from anywhere with an internet connection
You are responsible for physical security and backups	**Security**	Built-in security, high availability, and backup options
Fixed capacity (may be over-provisioned or under-utilized)	**Flexibility**	Flexible and adaptive to changing business needs
Best for stable, predictable workloads	**Ideal Use Case**	Ideal for modern, dynamic and global businesses

Both approaches can deliver great results.
Cloud computing gives you more flexibility, lower upfront cost, and faster innovation.

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Cloud Service Models

Cloud services are delivered in different ways based on how much control you need. The three main cloud service models are IaaS, PaaS and SaaS.

Same Cloud. Different Levels of Control.

IDEAS
APPLICATIONS
BUSINESS VALUE

Infrastructure as a Service (IaaS)

Get the infrastructure you need, when you need it.

- Provides virtualized computing resources over the internet.
- You manage the operating system, applications and data.
- Offers maximum flexibility and control.
- Ideal for companies that want to build and manage their own environments.

Examples

- Amazon EC2
- Virtual Machines
- Google Cloud Compute Engine

Use Cases
Web hosting, custom applications, development and test environments, disaster recovery.

Platform as a Service (PaaS)

Focus on building your application, not the infrastructure.

- Provides a complete development and runtime environment.
- You focus on writing and deploying your applications.
- The cloud provider manages the infrastructure, operating systems, middleware and more.
- Speeds up development and reduces operational complexity.

Examples

- AWS Elastic Beanstalk
- Azure App Service
- Google Cloud App Engine

Use Cases
Web and mobile application development, APIs, microservices, data analytics applications.

Software as a Service (SaaS)

Use ready-to-go software over the internet.

- Provides ready-to-use software applications over the internet.
- You do not manage the infrastructure, platform or application.
- Access the software from any device with a browser.
- Ideal for individuals and businesses that want to use software without worrying about maintenance.

Examples

- Google Workspace (Gmail, Drive, Docs)
- Microsoft 365 (Word, Excel, Teams)
- Salesforce (CRM)

Use Cases
Email, office productivity, customer relationship management, project management and more.

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Cloud Deployment Models

Cloud deployment models define where the cloud infrastructure is hosted and how it is made available to users. Each model has unique features and is suited for different business needs.

Same Cloud Technology.
Different Deployment Options.
Choose what fits your needs.

Public Cloud

Shared infrastructure, open to everyone.

Access over the internet

- Owned and managed by a cloud provider
- Shared infrastructure (multi-tenant)
- Accessible over the internet
- Scalable and cost-effective
- No need to manage physical hardware

Common Use Cases

- Web and mobile applications
- Startups and growing businesses
- Development and testing environments
- Applications with variable workloads

Private Cloud

Dedicated infrastructure for one organization.

- Dedicated to a single organization
- Owned and managed by the organization or a third-party provider
- Greater control and customization
- Higher security and compliance
- Typically higher cost than public cloud

Common Use Cases

- Businesses with strict regulatory requirements
- Sensitive data and critical applications
- Organizations that need full control
- Large enterprises

Hybrid Cloud

A combination of public and private clouds.

Connect. Integrate. Get the Best of Both.

- Combination of private and public clouds
- Workloads can move between environments
- Offers flexibility, scalability and control
- Helps optimize cost and performance
- Suitable for businesses with diverse needs

Common Use Cases

- Businesses with fluctuating workloads
- Data that needs to stay private with non-sensitive workloads in the public cloud
- Disaster recovery and backup
- Gradual migration to the cloud

Multi-Cloud

Use two or more cloud providers.

Multiple Clouds. More Possibilities.

- Uses services from two or more cloud providers
- Avoids vendor lock-in
- Choose the best service from each provider
- Improves resilience and reliability
- Can be more complex to manage

Common Use Cases

- Businesses that want to avoid vendor lock-in
- Using best-in-class services from different providers
- Global businesses with diverse requirements
- Improved redundancy and availability

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Benefits of Cloud Computing

Cloud computing allows individuals and organisations to access computing resources over the internet, offering many advantages compared to traditional IT infrastructure.

Access.
Scale.
Innovate.
Anywhere.

Cost Savings
Pay only for what you use. No need for large upfront investment in hardware.

Scalability
Easily increase or decrease resources based on your needs.

Flexibility
Access resources anytime, anywhere, from any device.

Faster Innovation
Quickly deploy new applications and ideas without waiting for physical infrastructure.

High Availability
Reliable and resilient infrastructure with built-in redundancy.

Improved Collaboration
Teams can work together in real time, from different locations.

Better Security
Advanced security features, regular updates and compliance support from cloud providers.

Environmental Sustainability
More efficient use of resources, which helps reduce energy consumption and carbon footprint.

Cloud computing empowers businesses and individuals to do more, with less cost, more speed and greater flexibility.

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BRIGHTER POSSIBILITIES.

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Introduction to AWS

Amazon Web Services (AWS) is the world's leading cloud computing platform, providing a wide range of on-demand services such as compute, storage, databases, networking, security, analytics, AI and more over the internet.

With AWS, you can build, deploy and scale applications globally without the need to own or manage physical servers.

aws

Build Without Limits

SCALABLE | SECURE | GLOBAL

On-Demand
Access resources when you need them

Cost-Effective
Pay only for what you use

Scalable
Easily scale up or down

Secure
Built-in security and compliance

Innovative
Access the latest technologies (AI, ML, analytics and more)

Global Reach
Deploy applications in multiple regions worldwide

> "From startups to global enterprises, AWS provides the building blocks for innovation."

Popular AWS Services

A wide range of services to meet different needs.

Amazon EC2
Virtual servers in the cloud

Amazon S3
Scalable object storage

Amazon RDS
Managed databases

Amazon VPC
Isolated cloud networks

AWS Lambda
Run code without managing servers

Amazon CloudWatch
Monitoring and logging

Amazon SageMaker
Build and deploy machine learning models

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AWS Global Infrastructure

AWS operates a global network of data centres that deliver compute, storage, databases and other services to customers around the world with high availability, low latency and scalability.

34+ Regions
Across the world

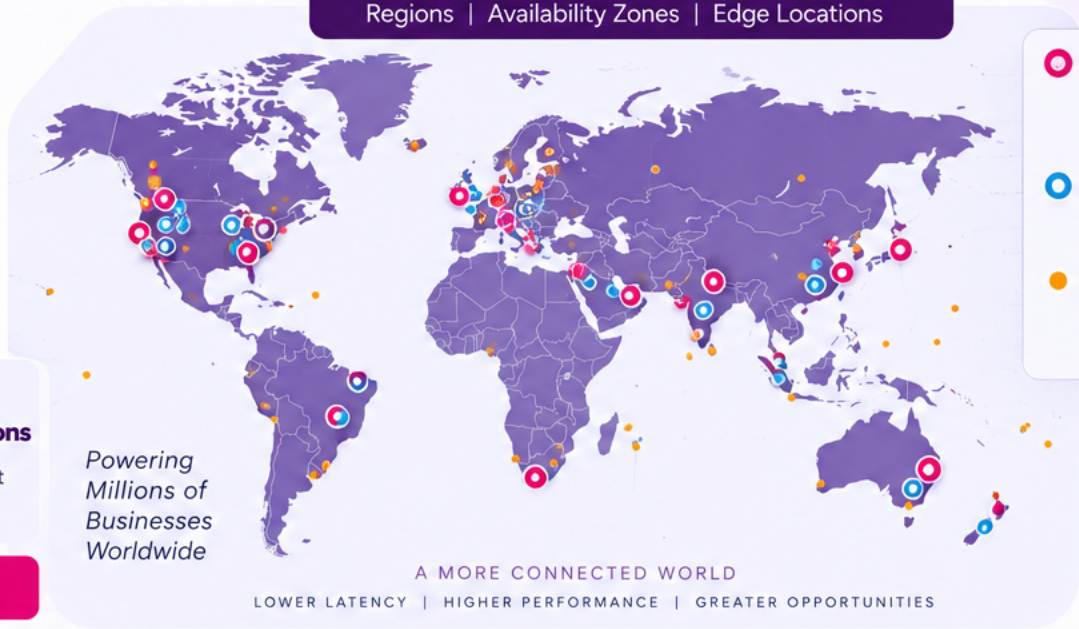
108+ Availability Zones
Multiple data centres per region

400+ Edge Locations
For fast content delivery

A global infrastructure. Closer to your users. Built for growth.

AWS Global Infrastructure

Regions | Availability Zones | Edge Locations



Region
A geographic area with multiple Availability Zones

Availability Zone
One or more data centres

Edge Location
Used by CloudFront to deliver content closer to users

Powering Millions of Businesses Worldwide

A MORE CONNECTED WORLD

LOWER LATENCY | HIGHER PERFORMANCE | GREATER OPPORTUNITIES

Build
Innovate
Scale
Anywhere

Regions

A Region is a physical location in the world that consists of multiple, isolated data centres (Availability Zones). Each Region is designed to be independent from other Regions.

Example: Africa (Cape Town), Europe (London), US (N. Virginia), Asia (Singapore).

Availability Zones (AZs)

An Availability Zone is one or more data centres with independent power, cooling and networking within a Region. AZs are connected with high-speed, low-latency networks, allowing you to build highly available applications.

Example: A Region like London has 3 or more Availability Zones (e.g. eu-west-2a, 2b, 2c).

Edge Locations

Edge Locations are used by Amazon CloudFront to cache content closer to users. This reduces latency and improves the speed and performance of your applications, websites and media content.

Example: Over 400+ edge locations worldwide in cities close to your users.

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AWS Shared Responsibility Model

Security and compliance is a shared responsibility between you (the customer) and AWS.

AWS is responsible for securing the cloud (infrastructure), while you are responsible for securing what you put in the cloud (your data, applications and configurations).

> "AWS provides a secure cloud foundation. You are responsible for how you use it.

Understand
Know what AWS manages and what you manage.

Configure
Set up your environment securely.

Stay Secure
Follow best practices and monitor regularly.

Customer Responsibility Security IN the cloud (You manage these)	AWS Responsibility Security OF the cloud (AWS manages these)
Data Your data, encryption and classification	**Physical Infrastructure** Data centres, hardware and facilities
Applications Your applications and code	**Global Infrastructure** Regions, Availability Zones, edge locations
Operating Systems OS configuration, patching and hardening	**Compute, Storage and Databases** EC2, S3, RDS and other services
Network Configuration Firewalls, security groups, routing	**Networking** Physical networking and service availability
Identity and Access Management Users, roles, permissions and policies	**Hypervisor** Virtualization layer
Security Configuration Monitoring, logging and security tools	**AWS Managed Services** Availability, scalability, and resilience of AWS services

A more secure cloud together

AWS and customers work together to keep data, applications and infrastructure secure, resilient and compliant.

Shared responsibility. Greater security. Bigger possibilities.

SAME CLOUD.
SHARED SECURITY.
BRIGHTER OPPORTUNITIES.

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AWS Identity and Access Management (IAM)

aws | Secure Access for Your AWS Resources

> "Give the right people the right access to the right resources."

AWS Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources. With IAM, you can create and manage users, groups, roles and permissions.

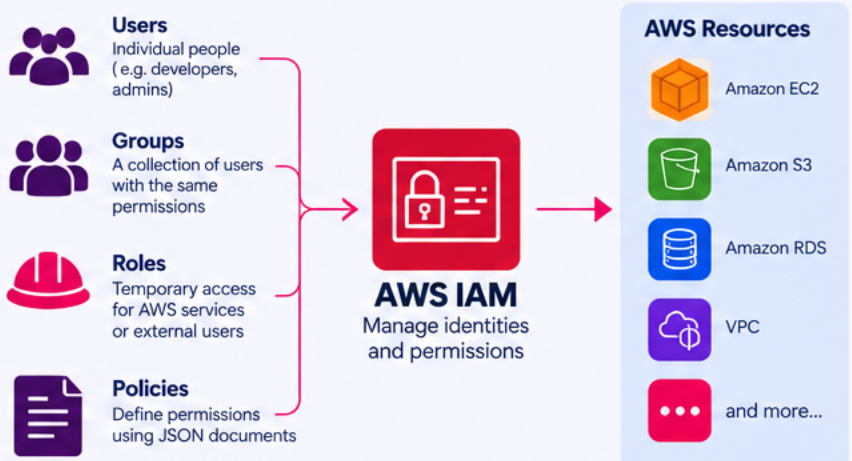
Manage Access
Control who can access AWS resources and what they can do.

Improve Security
Use least privilege access and multi-factor authentication (MFA).

Simplify Compliance
Monitor and audit access with detailed logging.

Flexible Access
Use roles for temporary access and integrate with external identities (e.g. your company directory).

How AWS IAM Works



Key Features of AWS IAM

- Create and manage IAM users, groups and roles
- Define permissions with IAM policies (JSON)
- Use multi-factor authentication (MFA)
- Implement least privilege access
- Integrate with AWS services and external identity providers (e.g. Active Directory, SAML, OpenID Connect)
- Manage temporary access with roles
- Monitor and audit activity with AWS CloudTrail
- Use permission boundaries and IAM conditions
- Support for cross-account access
- Help meet compliance and regulatory requirements

Common Use Cases

Team Access
Allow your team to access only the resources they need.

Application Access
Give applications and services permissions using roles.

Third-Party Access
Provide secure access to external partners or vendors.

Compliance and Auditing
Track and review access for security and regulatory requirements.

Best Practice

- Use the principle of least privilege
- Enable MFA for all users
- Use IAM roles instead of long-term access keys
- Regularly review and remove unused permissions
- Monitor access with AWS CloudTrail

Result

A more secure, compliant and well-managed AWS environment.

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Permissions and Authorization in AWS

Permissions define what actions are allowed, and authorization is the process of checking those permissions when a request is made to an AWS resource.

> "The right permissions to the right people at the right time."

Permissions

Permissions are rules that specify what actions a user, group or role is allowed or denied to perform on AWS resources.

Authorization

Authorization is the process of evaluating permissions to decide whether a request to access a resource should be allowed or denied.

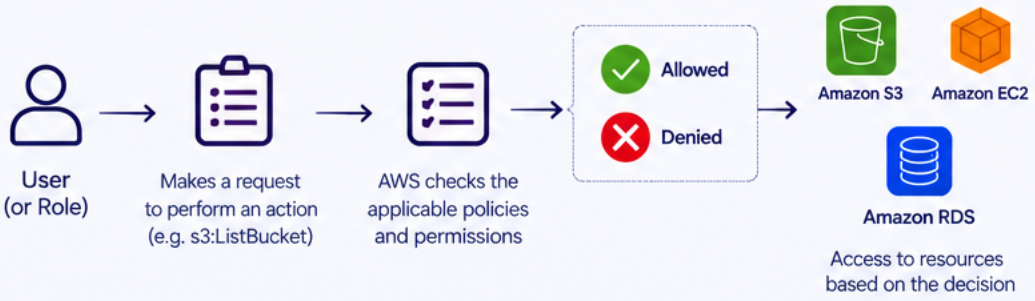
Example

A user with permission to s3:GetObject is authorized to read files from a specific S3 bucket. If the permission is not present, the request is denied.

Authentication vs Authorization

Authentication	Proves who you are (e.g. IAM user, MFA)
Authorization	Determines what you can do (permissions)

How It Works



IAM Policy Example

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::my-bucket",
        "arn:aws:s3:::my-bucket/*"
      ]
    }
  ]
}
```

Types of Permissions in AWS

Identity-based policies
Attached to IAM users, groups or roles. Define what actions are allowed or denied.

Resource-based policies
Attached to resources (e.g. S3 bucket). Define who can access the resource.

Permissions boundaries
Set the maximum permissions that an identity (user or role) can have.

Service Control Policies (SCPs)
Used in AWS Organizations to set guardrails for accounts.

Best Practices

- Follow the principle of least privilege
- Use IAM roles instead of long-term access keys
- Use condition keys to add extra restrictions
- Regularly review and remove unused permissions
- Use AWS CloudTrail to monitor API activity

Common Condition Keys

- **aws:SourceIp** – Restrict access to specific IP addresses
- **aws:MultiFactorAuthPresent** – Require MFA
- **aws:RequestedRegion** – Allow access only from specific regions
- **aws:SecureTransport** – Require HTTPS
- **s3:prefix** – Restrict access to specific folder paths in S3

Result

Proper permissions and authorization help you keep your AWS environment secure, compliant and well-managed.

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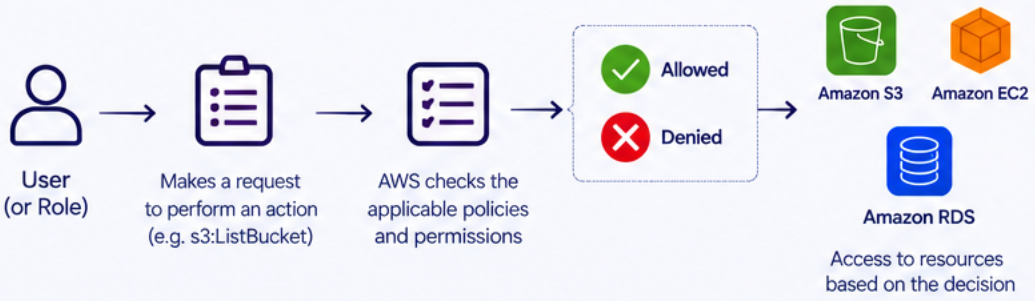

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Amazon EC2

Scalable virtual servers in the cloud

Amazon Elastic Compute Cloud (EC2) is a web service from AWS that provides secure, resizable compute capacity in the cloud. With EC2, you can launch virtual servers (**instances**), run your applications, and scale resources up or down based on your needs.

aws
Build
Deploy
Scale
On AWS

Reliable
Flexible
Cost-effective
Compute

Common Use Cases

- **Web Applications**
  Host websites and web apps.
- **Development and Testing**
  Set up environments quickly.
- **Databases**
  Run database servers (e.g. MySQL, PostgreSQL).
- **Big Data and Analytics**
  Process large datasets.
- **Machine Learning**
  Train and deploy ML models.
- **Enterprise Applications**
  Run business-critical workloads.
- **Disaster Recovery**
  Spin up infrastructure when needed.

Key Features

- **Scalable**
  Increase or decrease capacity in minutes.
- **Secure**
  Integrates with AWS IAM, security groups and key pairs.
- **Flexible**
  Wide range of instance types, OS and configurations.
- **Cost-effective**
  Pay only for what you use (with multiple pricing options).
- **Reliable**
  Built on AWS global infrastructure.
- **Integrated**
  Works seamlessly with other AWS services.

Instance Types (Examples)

Different instance types are optimized for different workloads.

General Purpose	Balanced compute, memory and networking (e.g. t3, m7i).
Compute Optimized	High-performance processors for compute-intensive workloads (e.g. c7i).
Memory Optimized	Designed for memory-intensive applications (e.g. r7i, x2idn).
Storage Optimized	High, sequential read/write access to large amounts of data (e.g. i4i, d3).
Accelerated Computing	Use GPUs or specialized hardware (e.g. p5, g5, inf2).

How EC2 Works

1. **Choose**
   Select an AMI and instance type.
2. **Configure**
   Set instance details (network, storage, security groups).
3. **Launch**
   Start your EC2 instance.
4. **Connect**
   Access via SSH (Linux) or RDP (Windows).
5. **Use and Scale**
   Run your applications and scale as needed.

Pricing Options

Choose the best pricing model for your workload.

- **On-Demand**
  Pay by the second, no long-term commitment.
- **Reserved Instances (RIs)**
  Lower cost for long-term workloads (1 or 3 years).
- **Savings Plans**
  Reduce costs with flexible pricing for consistent usage.
- **Spot Instances**
  Bid for unused capacity at lower prices (for fault-tolerant workloads).

Key Takeaway

Amazon EC2 gives you the power to run virtual servers in the cloud, with the flexibility to scale, the security to protect your data, and the cost options to fit your business needs.

Result

Build, deploy and scale applications faster with Amazon EC2.

> *"From a single server to global applications, EC2 makes it possible.*

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EC2 Instance Types

Choose the right instance type for your workload

Amazon EC2 offers a wide range of instance types, each optimized for different use cases in terms of compute, memory, storage, networking and acceleration.

aws

Right instance.
Right performance.
Right cost.

Compute
Store
Analyze
Innovate

General Purpose

T

T instances

A balanced mix of compute, memory and networking. Great for a wide range of applications.

Common use cases:

- Web servers
- Small and medium databases
- Development and testing
- Business applications

Compute Optimized

C

C instances

High-performance processors for compute-intensive workloads.

Common use cases:

- High-performance computing
- Batch processing
- Media encoding
- Scientific simulations

Memory Optimized

R

R instances

High memory-to-compute ratio for memory-intensive applications.

Common use cases:

- In-memory databases
- Big data analytics
- Real-time processing
- Enterprise applications (e.g. SAP, Oracle)

Storage Optimized

I

I instances

High, sequential read/write access to large amounts of data.

Common use cases:

- NoSQL databases
- Data warehousing
- Distributed file systems
- Log processing
- Big data workloads (e.g. Hadoop, Spark)

Accelerated Computing

P

P instances

Use GPUs, FPGAs or other accelerators for specialized workloads.

Common use cases:

- Machine learning and AI
- Deep learning
- High-performance computing
- Video transcoding
- Graphics rendering
- FPGA-based workloads

High Performance Computing

H

H instances

Optimized for tightly coupled, high-performance workloads with low latency networking.

Common use cases:

- HPC clusters
- Computational fluid dynamics
- Financial modeling
- Weather forecasting
- Large-scale simulations

Instance Type Naming Convention

m7i.large

- **m** — Instance family (m = general purpose)
- **7** — Generation (7)
- **i** — Processor (i = Intel, a = AMD, g = Graviton)
- **large** — Size (large)

Example Instance Types

Instance Type	Category	Description
t3.micro	General Purpose	Low cost, good for small workloads and learning
c7g.xlarge	Compute Optimized	Graviton processor, high compute performance
r7i.xlarge	Memory Optimized	High memory for demanding applications
i4i.xlarge	Storage Optimized	High I/O for storage-heavy workloads
p5.2xlarge	Accelerated Computing	NVIDIA GPU for AI/ML workloads
hpc7g.16xlarge	High Performance Computing	High performance with low latency networking

> **Different workloads. Different needs. The right instance type helps you get the best performance and value.**

Key Takeaway

Understand your workload requirements and choose the instance type that gives you the right balance of performance and cost.

Result

Right instance. Better performance.
Lower cost. Greater possibilities.

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Amazon Machine Images (AMI)

A pre-configured template for your EC2 instances

An Amazon Machine Image (AMI) is a template that contains the operating system, application software, configurations and settings required to launch an EC2 instance. You can use an AMI to create multiple instances with the same setup.

aws | Build Deploy Scale On AWS

Amazon Machine Image

> "Create once. Launch many. Stay consistent."

What Does an AMI Include?

Operating System
Linux, Windows or other OS.

Application Software
Pre-installed applications (e.g. web server, database, etc.).

Configurations
Settings, environment variables and custom configurations.

Block Device Mapping
Information about the storage (volumes) to attach.

Permissions
Controls who can use the AMI (e.g. private, public or shared).

How an AMI Works

Create or Choose an AMI
Use a pre-built AMI (AWS or custom) or create your own.

→

AMI Template
Contains OS, applications, configurations and settings.

→

Launch EC2 Instance
Use the AMI to launch one or more EC2 instances.

→

Multiple Instances
Launch multiple instances with the same configuration.

Types of AMIs

AWS Provided AMIs
Official AMIs from AWS (e.g. Amazon Linux, Ubuntu, Windows Server).

Community AMIs
Created and shared by the AWS community (e.g. pre-configured applications).

Custom AMIs
Created by you, based on your own instances (with your configurations and applications).

Common Use Cases

Web Servers
Pre-configured with Apache, Nginx, etc.

Application Servers
Pre-configured with application runtime (e.g. Java, Node.js).

Development and Testing
Create consistent environments for your team.

Backup and Recovery
Create an AMI of a configured instance for disaster recovery.

Standardization
Ensure all instances use the same, tested configuration.

How to Create a Custom AMI

1. **Launch and configure an EC2 instance**
   Install and configure your OS and applications.
2. **Prepare the instance**
   Remove temporary files and sensitive data.
3. **Create an AMI**
   From the EC2 console, choose "Create image".
4. **Add a description and tags**
   Give it a meaningful name and details.
5. **Launch new instances**
   Use your AMI to launch instances whenever needed.

Best Practices

- Keep AMIs up to date with latest security patches.
- Remove sensitive information before creating an AMI.
- Use descriptive names and tags.
- Store AMIs in multiple regions for disaster recovery.
- Use IAM permissions to control access to your AMIs.
- Regularly review and clean up unused AMIs to reduce costs.

Key Takeaway
An AMI gives you a consistent, reusable and scalable way to launch EC2 instances. It saves time, improves reliability and helps you maintain standard environments.

Result
Faster deployments. Consistent environments. Greater efficiency.

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EC2 Key Pairs

Secure access to your EC2 instances

An EC2 key pair is a set of cryptographic keys (a public key and a private key) used to securely connect to your EC2 Linux instances without a password.

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Your key.
Your access.
Keep it safe.

What is a Key Pair?

A key pair consists of:

- **Public key** – stored by AWS and used to encrypt the connection.
- **Private key** – downloaded to your computer and used to decrypt the connection when you log in to your instance.

Why Use Key Pairs?

- Provides secure, password-less access.
- Helps prevent unauthorized access.
- Required for connecting to Linux EC2 instances using SSH.
- You keep control of the private key.

Key Points to Remember

- You can only download the private key once when you create the key pair.
- AWS does not store your private key.
- If you lose the private key, you cannot access your instance (you will need to create a new instance or reset access).
- Key pairs are region-specific.

How Key Pairs Work

Your Computer (Private Key)
Uses the private key to decrypt and establish a secure connection (SSH).

Encrypted Connection (SSH)

EC2 Instance (Public Key)
AWS stores the public key on your instance and uses it to encrypt the connection.

Steps to Create a Key Pair (in AWS Console)

1. Go to **EC2 > Key Pairs.**
2. Click **Create key pair.**
3. Provide a name for your key pair.
4. Choose the key pair type (RSA) and file format (.pem for Linux).
5. Click **Create key pair** and download the private key file to your computer.

Connect to Your EC2 Instance (Linux)

```
# Set the correct permissions on your private key
chmod 400 my-key.pem

# Connect using SSH
ssh -i my-key.pem ec2-user@<your-instance-public-ip>
```

Important
Keep your private key file safe and do not share it with anyone. If it is lost, you will not be able to access your instance.

Best Practices

- Keep your private key in a secure location.
- Do not share your private key.
- Use meaningful names for your key pairs.
- Rotate key pairs periodically.
- Delete unused key pairs.

Common File Formats

.pem	Used for Linux/Mac (OpenSSH)
.ppk	Used for Windows (PuTTY)

Result

Key pairs provide a simple and secure way to connect to your EC2 instances, helping you keep your AWS environment safe.

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AWS Security Groups

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Allow only what you need. Keep your AWS environment secure.

Act as a virtual firewall for your EC2 instances

A security group controls inbound and outbound traffic to your AWS resources, such as EC2 instances. It only allows the traffic you specify and blocks everything else by default.

Key Points

- Acts as a virtual firewall.
- Controls inbound and outbound traffic.
- Stateful (if you allow inbound traffic, return traffic is automatically allowed).
- Applied at the instance level (or to other AWS resources like RDS, ALB, etc.).
- You can have multiple security groups attached to one instance.

Rules

- **Inbound rules:** Control traffic coming into your instance.
- **Outbound rules:** Control traffic going out from your instance.
- Each rule specifies:
  - **Protocol** (TCP, UDP, ICMP, etc.)
  - **Port range** (e.g. 22, 80, 443)
  - **Source** (e.g. specific IP, another security group, or anywhere 0.0.0.0/0)

Default Behavior

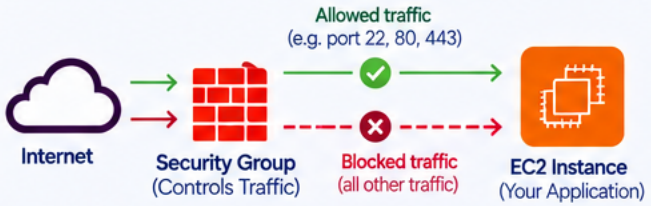
- All inbound traffic is blocked by default.
- All outbound traffic is allowed by default.
- You must add rules to allow the traffic you need.

Security groups are stateful – return traffic is automatically allowed.

Common Use Cases

- Allow SSH (port 22) from your IP address.
- Allow HTTP (port 80) and HTTPS (port 443) for web servers.
- Allow access from one security group to another (e.g. web server to database).
- Restrict access to specific IP ranges (e.g. your office network).
- Use different security groups for different environments (dev, test, prod).

How It Works



Example Inbound Rules

Type	Protocol	Port Range	Source	Purpose
SSH	TCP	22	203.0.113.10/32	Allow SSH from my IP address
HTTP	TCP	80	0.0.0.0/0	Allow web traffic from anywhere
HTTPS	TCP	443	0.0.0.0/0	Allow secure web traffic from anywhere
Custom TCP	TCP	3306	sg-12345678	Allow MySQL from another security group

Example Outbound Rules

Type	Protocol	Port Range	Destination	Purpose
All Traffic	All	All	0.0.0.0/0	Allow all outbound traffic (default)
HTTPS	TCP	443	0.0.0.0/0	Allow outbound HTTPS (e.g. updates)
DNS	UDP	53	0.0.0.0/0	Allow DNS resolution
Custom TCP	TCP	5432	sg-87654321	Allow PostgreSQL to another security group

Best Practices

- ✓ Use the principle of least privilege (allow only what you need).
- ✓ Restrict SSH access to your IP address (do not use 0.0.0.0/0).
- ✓ Use security groups to control access between tiers (e.g. web → app → database).
- ✓ Regularly review and remove unused rules.
- ✓ Use descriptive names and tags.
- ✓ Combine with other security measures (IAM, VPC, NACLs, etc.).

Common Mistakes

- ✗ Allowing SSH from anywhere (0.0.0.0/0).
- ✗ Opening unnecessary ports.
- ✗ Using the same security group for all resources.
- ✗ Forgetting outbound rules when needed.
- ✗ Not reviewing and cleaning up old rules.

Result

Properly configured security groups help you keep your applications secure while ensuring they remain accessible to authorized users.

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Launching an EC2 Instance

Step by step guide to launch a virtual server on AWS.

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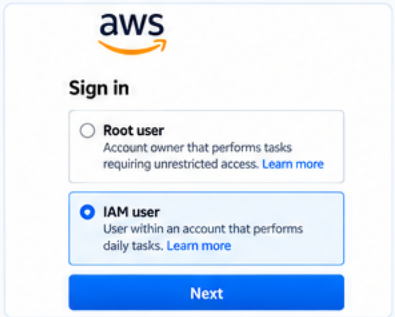
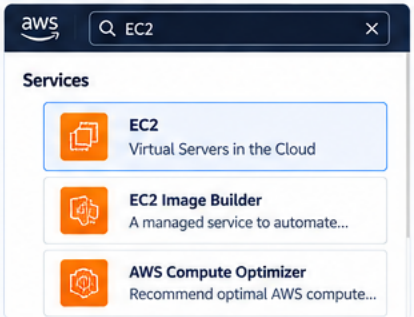
1 Sign in to AWS Console

Go to **https://aws.amazon.com** and sign in to your account.



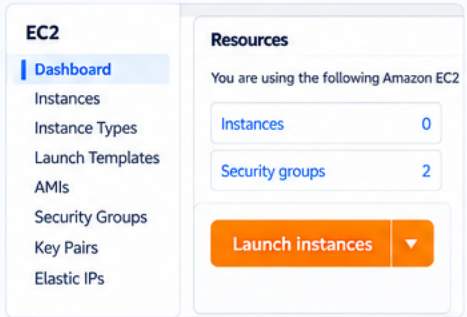
2 Go to EC2 Service

In the AWS Management Console, search for and open **EC2**.



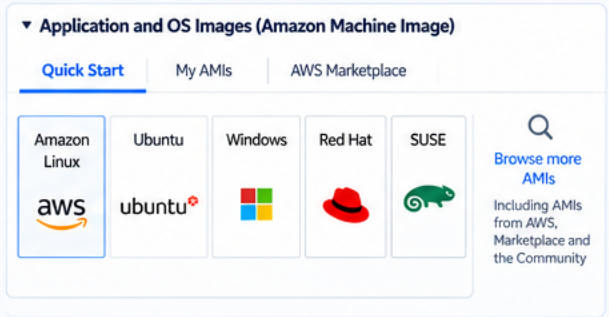
3 Click Launch Instances

On the EC2 dashboard, click **Launch instances**.



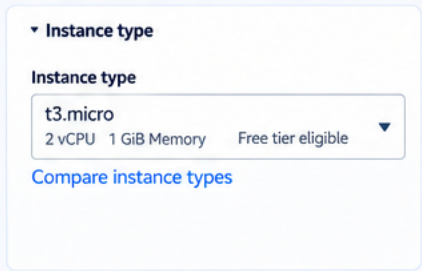
4 Choose an Amazon Machine Image (AMI)

Select an operating system (e.g. Amazon Linux, Ubuntu, Windows) or use your own AMI.



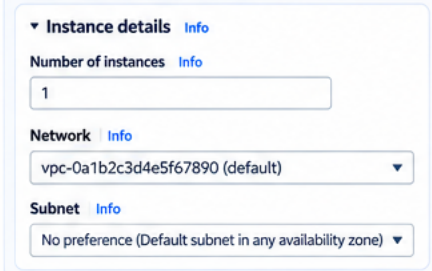
5 Choose an Instance Type

Select an instance type based on your compute, memory and other needs.



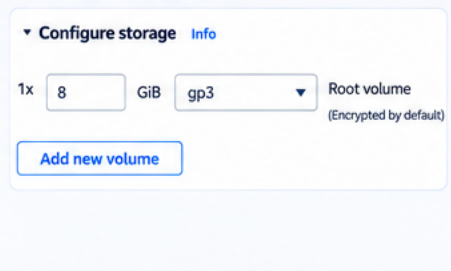
6 Configure Instance Details

Set the number of instances, network, subnet and other settings.



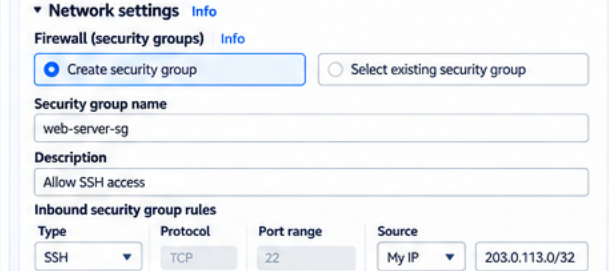
7 Add Storage

Specify the storage size (e.g. 8 GB) and volume type (e.g. gp3).



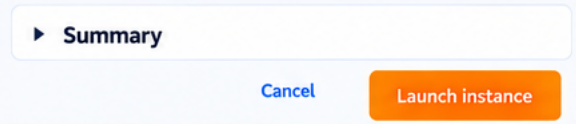
8 Configure Security Group

Create a new security group or select an existing one. Add rules to allow only the traffic you need (e.g. SSH).



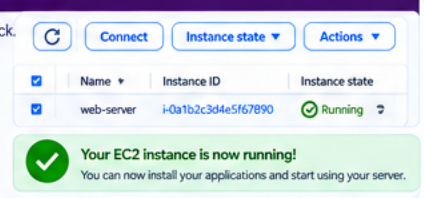
9 Review and Launch

Review your configuration and click Launch instance.



10 Connect to Your Instance

After launch, select your instance and click Connect. Use SSH (Linux) or RDP (Windows) to access it.



Tips for Success

- Use key pairs for secure access (do not use passwords).
- Follow the principle of least privilege in security groups.
- Use appropriate instance types for your workload.
- Keep your AMIs and OS up to date.
- Monitor your instance using Amazon CloudWatch.
- Stop or terminate unused instances to save costs.

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Connecting to a Linux EC2 Instance using SSH

Securely access your EC2 instance from your local computer

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1 Ensure You Have a Key Pair

When launching the EC2 instance, select an existing key pair or create a new one. This key pair is required for SSH access.

Key pair (login) Info

my-key-pair

Create new key pair

You will download a **.pem file** (e.g. **my-key-pair.pem**) when creating a new key pair. **Keep it safe.**

2 Check and Set Permissions on Your Key File

Your private key file should only be readable by you. Set the correct permissions using **chmod**.

```
# Go to the folder containing the key file
cd ~/Downloads

# Set the correct permissions
chmod 400 my-key-pair.pem

# Verify permissions (optional)
ls -l my-key-pair.pem
-r-------- 1 user staff 1675 Sep 15 10:30 my-key-pair.pem
```

If the permissions are too open, SSH will give an error.

3 Get Your EC2 Public IP Address

In the AWS EC2 console, select your instance and copy the **Public IPv4 address**.

Instance summary Info

Instance ID
i-0a1b2c3d4e5f67890

Public IPv4 address
54.123.45.67

Instance state
Running

Instance type
t3.micro

Your public IP address will be different from the one shown here.

4 Connect Using SSH

Use the **ssh** command with your key file and the EC2 public IP address.

```
ssh -i my-key-pair.pem
    ec2-user@54.123.45.67
```

Default usernames for common AMIs:

- Amazon Linux / Amazon Linux 2 → **ec2-user**
- Ubuntu → **ubuntu**
- RHEL / CentOS → **ec2-user**
- SUSE → **ec2-user**
- Debian → **admin**

5 Accept the Host Key (First Time)

The first time you connect, you will see a message like this:

```
The authenticity of host '54.123.45.67 (54.123.45.67)'
can't be established.
ED25519 key fingerprint is SHA256:....
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press **Enter**.

6 Successful Connection

If successful, you will see the Linux prompt.

```
Last login: Mon Sep 15 10:35:10 2026 from 203.0.113.10

       Amazon Linux 2023

[ec2-user@ip-172-31-12-34 ~]$
```

7 Basic Commands to Try

Once connected, try some basic commands.

```
[ec2-user@ip-172-31-12-34 ~]$ whoami
ec2-user
[ec2-user@ip-172-31-12-34 ~]$ uname -a
Linux ip-172-31-12-34 6.1.0-xxxxxxx x86_64
[ec2-user@ip-172-31-12-34 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvda1      8.0G  1.5G  6.5G  19% /
```

8 Disconnect When Done

Type **exit** to close the SSH session.

```
[ec2-user@ip-172-31-12-34 ~]$ exit
logout
Connection to 54.123.45.67 closed.
```

Key Takeaways

- Use a key pair for secure, password-less access.
- Set correct file permissions (chmod 400).
- Use the correct username for your AMI.
- Keep your key file safe and do not share it.

Common Issues

- Permission denied (publickey) → Check key file permissions.
- Connection timed out → Check security group allows SSH (port 22).
- Host key verification failed → Remove old entry from known_hosts.
- Wrong username → Use the correct default username for your AMI.

Result

- Securely connected to your Linux EC2 instance.
- Ready to install, configure and manage your applications.

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EC2 IP Addresses

Understand the different IP addresses used by your EC2 instances

Every EC2 instance gets IP addresses that allow it to communicate within the AWS network and with the internet. Understanding these IP addresses helps you manage access, security and connectivity.

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Types of EC2 IP Addresses

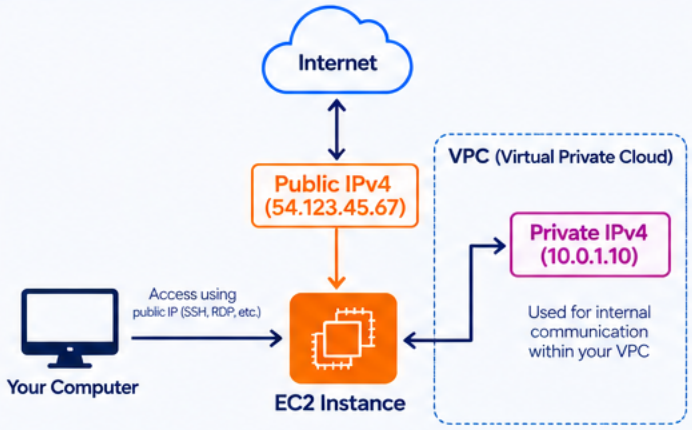
Private IPv4 Address
Used for internal communication within your VPC. Not accessible from the internet.

Public IPv4 Address
Used to access your instance from the internet. Can be auto-assigned or Elastic.

Elastic IP Address (EIP)
A static public IP that you can associate with your instance. Remains the same even if you stop and start the instance.

IPv6 Address
A publicly routable IPv6 address (available in supported regions). Useful for modern applications and dual-stack deployments.

How EC2 IP Addresses Work



Where to Find EC2 IP Addresses

1. **Go to the AWS Management Console**
   Open the EC2 service.
2. **Select Your Instance**
   Click on your instance from the list.
3. **View IP Addresses**
   In the instance details, you will see:
   - Public IPv4 address (if assigned)
   - Private IPv4 address
   - IPv6 address (if enabled)
   - Elastic IP (if associated)

Key Points to Remember

- A public IP is required to access your instance from the internet.
- A private IP is always assigned (even without a public IP).
- You can use an Elastic IP to keep the same public IP.
- If you stop an instance, the public IP changes (unless you use an Elastic IP).
- IPv6 is available in supported regions and can be used alongside IPv4.

Best Practices

- Use a security group to restrict access to your public IP.
- Use Elastic IP for services that require a fixed IP.
- Prefer private IP for internal communication between AWS resources.
- Remove public IP if it is not required (use a private subnet).
- Consider using a bastion host or AWS Systems Manager (Session Manager) instead of exposing public IPs.

Common Use Cases

- Access your EC2 instance using SSH (Linux).
- Access your EC2 instance using RDP (Windows).
- Host a web application (public IP or Elastic IP).
- Allow internal communication between EC2 instances using private IPs.
- Use Elastic IP for critical services.

Key Takeaway
EC2 instances use different types of IP addresses for different purposes. Choose the right IP configuration based on your security and connectivity requirements.

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EC2 User Data and Metadata

Automate configuration and access instance information easily

1 EC2 User Data

User data is a script that runs automatically when an EC2 instance is launched for the first time. It allows you to automate instance configuration, such as installing software, updating packages, and starting services.

- **Runs once at launch** — Executed during the first boot only.
- **Supports scripts** — Commonly used with shell scripts (Bash), PowerShell or cloud-init.
- **Automates setup** — Install packages, configure applications, start services, etc.
- **Saves time** — Helps you launch fully configured instances.

Example: Linux User Data (Bash)

```
#!/bin/bash
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx
echo "<h1>Welcome to Tech365 EC2</h1>" > /usr/share/nginx/html/index.html
```

How to Add User Data When Launching an EC2 Instance

1. In the AWS EC2 console, click **Launch instance**.
2. Scroll down to the **Advanced details** section.
3. Expand **User data** and paste your script.
4. Launch the instance. The script will run on first boot.

Common Use Cases

Install and configure web servers (e.g. Nginx, Apache).

- Install application dependencies.
- Pull code from GitHub.
- Configure environment variables.
- Join the instance to a domain.
- Set up monitoring or logging agents.

2 EC2 Metadata

Instance metadata provides information about your EC2 instance and its environment. It is available from within the instance and can be accessed using a special IP address (169.254.169.254). Metadata is useful for retrieving instance details, IAM role credentials, network information and more.

- **Accessible from Inside the instance** — Use the link-local IP 169.254.169.254.
- **Provides instance information** — Instance ID, region, IP addresses, IAM role, etc.
- **Supports IAM credentials** — Retrieve temporary credentials for the instance role.
- **Version 2 (IMDSv2)** — More secure. Use session tokens to access metadata.

Example: Accessing Metadata (Linux)

```
# Get the instance ID
curl http://169.254.169.254/latest/meta-data/instance-id

# Get the instance type
curl http://169.254.169.254/latest/meta-data/instance-type

# Get the public IPv4 address
curl http://169.254.169.254/latest/meta-data/public-ipv4

# Get IAM role name
curl http://169.254.169.254/latest/meta-data/iam/security-credentials/
```

Using IMDSv2 (Recommended)

1. Get a session token (valid for 6 hours).

```
TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
```

2. Use the token to access metadata.

```
curl -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/instance-id
```

Key Takeaways

- Use user data to automate instance setup at launch.
- Use instance metadata to retrieve information from inside the instance.
- Prefer IMDSv2 for better security.
- Both features help you build more automated and scalable solutions on AWS.

Important Notes

- User data runs only once, at the first boot.
- Metadata is accessible only from within the instance (not from the internet).
- Use IMDSv2 to protect against SSRF (Server-Side Request Forgery) attacks.
- Do not store sensitive data directly in user data scripts.

Result

- Faster and consistent instance deployment.
- Easy access to instance information.
- More secure and automated infrastructure management.

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EC2 Pricing Models

Choose the right pricing model for your workload and budget

Amazon EC2 offers flexible pricing options to meet different needs, from short-term testing to long-term, predictable workloads. Understanding these options helps you optimize cost and get the best value from AWS.

1 On-Demand Instances

Pay for what you use, by the hour or second

On-Demand instances provide flexibility to run EC2 instances without a long-term commitment. You pay only for the compute capacity you use.

Key Benefits

- No long-term commitment
- Pay by the hour or second
- Quick and easy to start
- Ideal for variable workloads

Common Use Cases

- Development and testing
- Short-term projects
- Unpredictable workloads
- Applications with changing demand

2 Reserved Instances (RIs)

Commit and save

Reserved Instances let you commit to using an instance for 1 or 3 years in exchange for a significant discount compared to On-Demand pricing.

Key Benefits

- Lower cost (up to 72% savings)
- Predictable pricing
- Multiple payment options (All Upfront, Partial Upfront, No Upfront)
- Suitable for steady-state workloads

Common Use Cases

- Production applications
- Long-term workloads
- Databases and enterprise applications
- Workloads with predictable usage

3 Savings Plans

Flexible commitment, bigger savings

Savings Plans offer lower prices than On-Demand in exchange for a commitment to a consistent amount of compute usage (measured in $/hour) for 1 or 3 years. Provides more flexibility than Reserved Instances.

Key Benefits

- Lower cost (up to 72% savings)
- Flexible across instance types, sizes, and operating systems
- Works with EC2, Fargate and Lambda
- Ideal for changing workloads

Common Use Cases

- Organizations with dynamic workloads
- Mixed instance types
- Growing applications
- When you want flexibility with savings

4 Spot Instances

Take advantage of unused capacity

Spot Instances let you use unused EC2 capacity at greatly reduced prices (up to 90% off On-Demand). Instances can be interrupted when AWS needs the capacity back.

Key Benefits

- Very low cost (up to 90% savings)
- Ideal for fault-tolerant workloads
- Access to large compute capacity
- Great for batch processing

Common Use Cases

- Batch processing
- Data analytics
- Image and video rendering
- Machine learning training
- Fault-tolerant and flexible workloads

5 Dedicated Hosts and Dedicated Instances

Physical servers for your specific needs

Dedicated Hosts give you an entire physical server, while Dedicated Instances run on hardware that is dedicated to you (not shared with other customers).

Key Benefits

- More control over hardware
- Helps meet compliance and regulatory requirements
- Use existing server-bound licenses
- Dedicated Instances offer dedicated capacity without purchasing the entire server

Common Use Cases

- Regulated industries (e.g. finance, healthcare)
- Applications with server-bound licenses
- High security or compliance requirements
- Migrating from on-premises environments

Comparison at a Glance

Pricing Model	Cost	Commitment	Flexibility	Best For
On-Demand	Highest	None	High	Short-term, variable workloads
Reserved Instances	Lower (up to 72%)	1 or 3 years	Low	Steady-state, long-term workloads
Savings Plans	Lower (up to 72%)	1 or 3 years	Higher	Flexible, changing workloads
Spot Instances	Lowest (up to 90%)	None	High (but interruptible)	Batch, fault-tolerant workloads
Dedicated Hosts	Highest	1 or 3 years	Low	Compliance, server-bound licenses

Which Model Should You Choose?

- Use **On-Demand** for short-term and unpredictable needs.
- Use **Reserved Instances** for long-term, steady workloads.
- Use **Savings Plans** for flexible, cost-effective savings.
- Use **Spot Instances** for fault-tolerant, non-critical workloads.
- Use **Dedicated Hosts/Instances** for compliance and dedicated hardware needs.

Key Takeaway

There is no single best pricing model. The right choice depends on your workload requirements, budget, and flexibility needs. Many organizations use a combination of these models to optimize cost and performance.